DataSynth

Enterprise Synthetic Data Platform

High-Performance Generation for Accounting, Audit & ML

Rust core • Python 1.1 • REST/gRPC API • Desktop UI • Helm/Docker

Executive Overview

Version 0.6.0

Deterministic, privacy-preserving synthetic data generation
for enterprise finance, compliance testing, and machine learning.

Built with Rust & Python • 15 modular crates • 100 K+ entries/sec

Open-source (Apache-2.0) • Commercial license available

Contact: michael.ivertowski@ch.ey.com

Contents

1 Executive Summary **3**
1.1 At a Glance . . . 3

2 Platform Capabilities **4**
2.1 Enterprise Accounting Simulation . . . 4
2.2 Enterprise Process Chains . . . 5
2.3 Banking, KYC & AML . . . 5
2.4 Audit Simulation . . . 5
2.5 COSO 2013 Internal Control Framework . . . 6
2.6 Accounting & Audit Standards . . . 6
2.7 Process Mining (OCEL 2.0 & XES 2.0) . . . 6

3 AI-Augmented Generation **7**
3.1 LLM-Augmented Metadata Enrichment . . . 7
3.2 Diffusion Model Integration . . . 7
3.3 Causal & Counterfactual Generation . . . 7

4 Machine Learning & Analytics **8**
4.1 ACFE-Aligned Fraud Taxonomy . . . 8
4.2 Advanced Fraud Modeling . . . 8
4.3 Data Quality Variations . . . 8
4.4 Enterprise Interconnectivity . . . 9
4.5 Pattern & Process Drift . . . 9
4.6 Graph & Network Export . . . 9

5 Privacy & Fingerprinting **10**
5.1 Privacy-Preserving Fingerprint Extraction . . . 10
5.2 Privacy Levels . . . 10
5.3 Federated Fingerprinting & Certificates . . . 10
5.4 Formal DP Composition . . . 10
5.5 Fidelity Evaluation . . . 11

6 Architecture **11**
6.1 Layered Crate Architecture . . . 11
6.2 Production-Grade Infrastructure . . . 12

7 Industry Presets **13**

8 Statistical Distribution Framework **13**

9 Use Cases **14**

10 Evaluation & Auto-Tuning **14**
10.1 ML Benchmarks . . . 15
10.2 Quality Gate Engine . . . 15

11 Compliance & Governance **15**
11.1 EU AI Act . . . 15
11.2 Regulatory Compliance Documentation . . . 16

12 DevOps & Deployment **16**
12.1 CI/CD Pipeline 16
12.2 Deployment Options 16
12.3 Ecosystem Integrations 16

13 Getting Started **18**
13.1 Quick Start (CLI) 18
13.2 Quick Start (Python) 18
13.3 Server Mode 18

1 Executive Summary

DataSynth is a high-performance synthetic data platform purpose-built for enterprise accounting, audit analytics, and machine learning. Written in Rust for maximum throughput and memory safety — with a full Python wrapper for data-science workflows — it generates realistic, internally coherent financial data that satisfies the statistical and structural properties of real-world enterprise resource planning (ERP) systems.

Why DataSynth?

- **No real data required** — eliminates privacy, regulatory, and procurement barriers to analytics development.
- **Full auditability** — deterministic, seeded generation means every dataset is perfectly reproducible.
- **ML-ready from day one** — ground-truth labels for fraud, anomalies, and data quality ship alongside every record.
- **Domain depth** — covers the full accounting lifecycle: journal entries, document flows, subledgers, FX, intercompany, period close, and banking/AML.
- **Empirically grounded** — statistical distributions for journal entry line items, amount patterns, and temporal volumes are calibrated against empirical research conducted on real-world enterprise datasets, ensuring synthetic output mirrors the structural properties of production ERP data.

1.1 At a Glance

100 K+ entries/sec (single thread)	**60+** ACFE-aligned fraud schemes	**10** industry presets
15 modular Rust crates	**50+** red flag patterns	**4** privacy levels

2 Platform Capabilities

2.1 Enterprise Accounting Simulation

DataSynth generates a complete, internally consistent accounting universe:

Domain	Capabilities
Journal Entries	Balanced debits/credits, Benford-compliant amounts, configurable line-item distributions, SAP ACDOCA format export.
Master Data	Vendors, customers, materials, fixed assets, employees with hierarchies, payment terms, credit ratings, and intercompany flags.
Document Flows	Full Procure-to-Pay (PO $\rightarrow$ GR $\rightarrow$ Invoice $\rightarrow$ Payment) and Order-to-Cash (SO $\rightarrow$ Delivery $\rightarrow$ Invoice $\rightarrow$ Receipt) with three-way match validation and contract price enforcement.
Intercompany	Matched IC journal entry pairs, transfer pricing (Cost-Plus, Resale-Minus, CUP), and consolidation elimination entries.
Subledgers	AR/AP open items and aging, fixed asset register with IRS MACRS depreciation tables (3–20 yr recovery periods), inventory positions and movements, GL-to-subledger reconciliation.
FX & Translation	Ornstein–Uhlenbeck exchange rate process, multi-currency trial balance translation (current-rate, temporal, monetary/non-monetary methods with account-type classification), currency translation adjustment entries.
Period Close	Month-end accruals, depreciation runs, year-end closing entries, fiscal period status tracking.

2.2 Enterprise Process Chains

New in v0.6.0 — five end-to-end process chains with full referential integrity, temporal coherence, and anomaly injection:

Chain	Capabilities
Source-to-Pay	Spend analysis, sourcing projects, supplier qualification, RFx/bid evaluation, procurement contracts, catalog management, supplier scorecards — with contract-governed PO generation and maverick spend detection.
Hire-to-Retire	Payroll runs with tax/deduction calculations, time entries with overtime modeling, expense reports with policy violation detection, and payroll journal entry generation.
Manufacturing	Production orders with BOM explosion, routing operations, component issues, finished goods receipts, WIP costing (material/labor/overhead), and variance settlement.
Bank Reconciliation	Statement line matching against cleared payments, outstanding checks, deposits in transit, bank fees, and zero-net-difference enforcement.
Financial Reporting	Balance sheet, income statement, cash flow statement (indirect method), and changes in equity — derived from trial balance with prior-period comparatives, management KPIs, and budget variance analysis.

Each chain defaults to `enabled: false` and integrates with the existing generation pipeline via new orchestrator phases. All new fields on existing structs use `Option<>` for backward compatibility.

2.3 Banking, KYC & AML

A dedicated banking module generates realistic transaction data for anti-money-laundering testing:

- **Customer personas**: Retail, Business, Trust profiles with full KYC envelopes (declared turnover, source of funds, geographic exposure, cash intensity).
- **AML typologies**: Structuring, funnel accounts, layering schemes, money mule networks, round-tripping, and adversarial spoofing for robustness testing.
- **Ground-truth labels**: Entity-level risk classifications, transaction-level labels, and investigation narratives.

2.4 Audit Simulation

Generates ISA-compliant audit artifacts:

- Engagement metadata with materiality thresholds (ISA 320).
- Workpapers per ISA 230, evidence per ISA 500.
- Risk assessments (ISA 315/330), findings (ISA 265), and professional judgment documentation (ISA 200).

2.5 COSO 2013 Internal Control Framework

Full integration with the COSO Internal Control-Integrated Framework:

- **5 Components**: Control Environment, Risk Assessment, Control Activities, Information & Communication, Monitoring Activities.
- **17 Principles**: Complete principle coverage with control mappings.
- **Control Scopes**: Entity-level, transaction-level, IT general, and IT application controls.
- **Maturity Levels**: 6-level model (Non-Existent through Optimized).
- **Export**: `coso_control_mapping.csv` with principle-level granularity.

2.6 Accounting & Audit Standards

Comprehensive standards support via the `datasynth-standards` crate:

Category	Standards Supported
Accounting (US GAAP)	ASC 606 (Revenue), ASC 842 (Leases — 5 bright-line classification tests), ASC 820 (Fair Value), ASC 360 (Impairment)
Accounting (IFRS)	IFRS 15 (Revenue), IFRS 16 (Leases — enhanced ROU measurement), IFRS 13 (Fair Value), IAS 36 (Impairment)
Audit Standards	34 ISA standards (ISA 200–720), 19+ PCAOB standards with ISA mapping
Regulatory	SOX Section 302 (Certifications), SOX Section 404 (ICFR Assessment), Deficiency Matrix (MW/SD classification)

Supports dual reporting mode (US GAAP + IFRS) with automatic framework difference reconciliation.

2.7 Process Mining (OCEL 2.0 & XES 2.0)

Object-Centric Event Logs track many-to-many relationships between business objects (orders, invoices, payments) and activities — enabling conformance checking and process variant analysis.

- **OCEL 2.0**: JSON and XML export with objects, events, and object relationships.
- **XES 2.0**: IEEE standard format for ProM, Celonis, Disco, and pm4py.
- **Reference Models**: Canonical P2P (9 activities), O2C (10 activities), R2R (11 activities), S2C (9 activities), H2R, and MFG process models for conformance checking.
- **30+ R2R/A2R/S2C Activities**: GL posting, FX revaluation, period close, depreciation, consolidation, sourcing, qualification, bidding, contract management.
- **Deterministic UUIDs**: FNV-1a hash-based IDs with type discriminators for reproducible event logs.

3 AI-Augmented Generation

New in v0.5.0 — three opt-in AI backends that enhance the rule-based core without sacrificing determinism or reproducibility.

3.1 LLM-Augmented Metadata Enrichment

An `LlmProvider` trait integrates external language models for contextual metadata enrichment while preserving deterministic fallback:

- **Provider trait**: `MockProvider` (deterministic, default) and `HttpProvider` (OpenAI, Anthropic, or custom API) implementations.
- **Enrichment targets**: Vendor names, transaction descriptions, anomaly explanations, and memo fields.
- **Natural language config**: Generate YAML configurations from plain English descriptions.
- **Response caching**: LRU cache with deterministic fallback ensures reproducibility even with stochastic providers.

3.2 Diffusion Model Integration

A statistical diffusion backend learns distribution shapes from fingerprints and blends them with rule-based outputs:

- **Noise schedules**: Cosine, linear, and sigmoid with configurable step count.
- **Hybrid mode**: Weighted blend of rule-based and diffusion-generated values (`diffusion_weight` 0.0–1.0).
- **Training pipeline**: `DiffusionTrainer` with epoch count, batch size, and learning rate configuration.

3.3 Causal & Counterfactual Generation

Structural causal models enable "what-if" scenario generation grounded in do-calculus:

- **CausalGraph**: DAG specification of causal relationships between entities (e.g., vendor risk → invoice amount → approval level).
- **StructuralCausalModel**: Functional causal model with noise terms and topological evaluation.
- **Interventions**: Do-calculus interventions for hypothetical scenarios ("what if vendor risk doubled?").
- **Counterfactuals**: Generate alternative versions of existing records under modified assumptions.
- **Templates**: Pre-built causal graphs for fraud detection, revenue impact analysis, and supply chain modeling.

4 Machine Learning & Analytics

4.1 ACFE-Aligned Fraud Taxonomy

DataSynth implements the ACFE (Association of Certified Fraud Examiners) "Report to the Nations" taxonomy with calibrated statistics:

ACFE Category	Prevalence	Scheme Types
Asset Misappropriation	86%	20+ cash schemes: skimming, larceny, shell company, ghost employee, billing fraud, expense reimbursement
Corruption	33%	Conflicts of interest, bribery, kickbacks, bid rigging, economic extortion
Financial Statement	10%	Revenue manipulation, expense timing, concealed liabilities, improper disclosures

Calibration: $117k median loss, 12-month median duration, detection method distribution (42% tips, 16% internal audit, 12% management review).

4.2 Advanced Fraud Modeling

- **Collusion & Conspiracy**: 9 ring types (EmployeePair, DepartmentRing, EmployeeVendor, VendorRing), 6 conspirator roles, defection modeling based on detection risk.
- **Management Override**: Executive-level fraud with override techniques, Fraud Triangle modeling (Pressure, Opportunity, Rationalization), concealment strategies.
- **Red Flag Patterns**: 40+ patterns with Bayesian probabilities, 6 categories (Vendor, Transaction, Timing, Approval, Document, Behavioral), realistic false positive rates.
- **Industry-Specific**: Manufacturing (yield manipulation, phantom production), Retail (sweethearting, refund fraud), Healthcare (upcoding, unbundling, phantom billing).
- **Procurement Fraud**: Bid rigging, phantom vendor contracts, split-to-avoid-threshold, conflict of interest, maverick spend, expired contract purchases, qualification bypass.
- **HR/Payroll Fraud**: Ghost employee payroll, payroll inflation, duplicate expense reports, fictitious expenses, split expense to avoid approval.

Anomaly injection is **entity-aware**: injection rates are adjusted per vendor, employee, and account based on contextual risk factors (e.g., new vendors, high-value accounts, employees with recent access changes). Every injected anomaly carries a `LabeledAnomaly` record with full metadata, enabling supervised and semi-supervised learning pipelines without manual labeling effort.

4.3 Data Quality Variations

Realistic data imperfections for training data-quality ML models:

- **Missing values**: MCAR, MAR, MNAR, and systematic patterns.
- **Format variations**: Date, amount, and identifier format diversity across regional conventions.
- **Duplicates**: Exact, near, and fuzzy duplicates.
- **Typos**: Keyboard-aware substitution, OCR errors, homophones.
- **Encoding issues**: Mojibake, BOM artifacts, HTML entity corruption.

4.4 Enterprise Interconnectivity

Realistic enterprise network modeling with configurable relationships:

- **Multi-Tier Vendor Networks**: 3-tier supply chain (Tier1/Tier2/Tier3), vendor clusters (ReliableStrategic 20%, StandardOperational 50%, Transactional 25%, Problematic 5%), lifecycle stages, quality scores, dependency analysis.
- **Customer Value Segmentation**: Enterprise (40% revenue/5% customers), MidMarket (35%/20%), SMB (20%/50%), Consumer (5%/25%), lifecycle stages, referral networks, corporate hierarchies.
- **Entity Relationship Graph**: 16 entity types, 26 relationship types, composite strength calculation from volume, count, duration, recency, and mutual connections.
- **Cross-Process Links**: P2P↔O2C linkage via inventory (GoodsReceipt→Delivery), intercompany bilateral links.

4.5 Pattern & Process Drift

Comprehensive drift modeling for realistic temporal evolution:

- **Organizational Events**: Acquisition (1.35x volume, 5% integration errors), divestiture, reorganization, leadership change, workforce reduction, merger.
- **Process Evolution**: Approval workflow changes, S-curve automation rollout, policy changes, control enhancements.
- **Technology Transitions**: ERP migration phases with error rate and processing time multipliers, duplicate/missing/format issues.
- **Behavioral Drift**: Vendor payment term extensions, customer delays during downturns, employee learning curves, year-end intensity.
- **Market Drift**: Economic cycles (sinusoidal, asymmetric, mean-reverting), industry-specific cycles, commodity drift, price shocks.
- **Ground Truth**: `LabeledDriftEvent` with event type, magnitude, and detection difficulty (Easy/Medium/Hard) for drift detection ML.

4.6 Graph & Network Export

Supported Graph Formats

PyTorch Geometric	`.pt` files with node features, edge index, edge attributes, labels, and train/val/test masks.
Neo4j	CSV node/edge files with Cypher import scripts.
DGL	Deep Graph Library format for GNN training.
RustGraph	Native Rust graph format with adjacency lists.
RustGraph Hyper	Multi-layer hypergraph export (entity, temporal, process layers with cross-layer edges).

All five exporters share a unified `CommonExportConfig` for consistent feature engineering, label selection, and train/val/test mask settings. Computed features include temporal signals (weekday, period-end flags), amount signals (log-amount, Benford probability, round-number flag), structural signals (line count, unique accounts), and one-hot categorical encodings.

5 Privacy & Fingerprinting

5.1 Privacy-Preserving Fingerprint Extraction

DataSynth can extract a statistical *fingerprint* from real data and synthesize new data that matches its properties — without ever copying individual records.

5.2 Privacy Levels

Level	Epsilon (ε)	k-Anon	Outlier %	Use Case
Minimal	5.0	3	99%	Low privacy, high utility
Standard	1.0	5	95%	Balanced (default)
High	0.5	10	90%	Sensitive environments
Maximum	0.1	20	85%	Maximum privacy guarantees

The privacy engine combines differential privacy (Laplace and Gaussian mechanisms), k-anonymity with rare-value suppression, and outlier winsorization. A full privacy audit trail records every decision and the cumulative epsilon budget spent.

5.3 Federated Fingerprinting & Certificates

New in v0.5.0 — distributed fingerprint extraction and cryptographic provenance:

- **Federated Coordinator**: Aggregate fingerprints from distributed data sources without centralizing raw data.
- **Synthetic Data Certificates**: `SyntheticDataCertificate` with cryptographic proof of differential privacy guarantees, generation metadata, and validation workflow.
- **Pareto Frontier**: Automated exploration of optimal privacy–utility trade-offs across epsilon values.

5.4 Formal DP Composition

New in v0.5.0 — advanced privacy accounting beyond naive ε-summation:

- **Rényi DP**: RDP curve tracking at α values 2–128 with conversion to (ε,δ)-DP.
- **zCDP**: Zero-Concentrated DP with additive ρ composition and tighter bounds.
- **Budget Manager**: Global budget tracking across extraction runs with composition-aware enforcement.
- **Privacy Evaluation**: Post-generation membership inference attack (MIA) testing, linkage attack assessment, and NIST SP 800-226 alignment.

5.5 Fidelity Evaluation

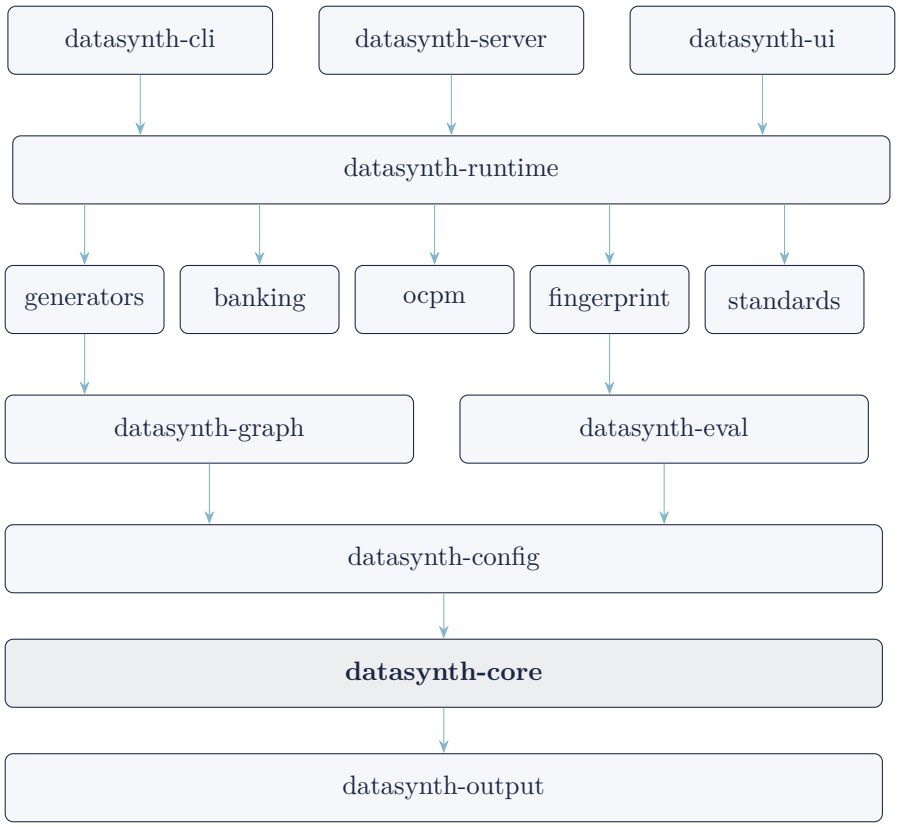
Per-column statistical distance metrics quantify how well synthetic data matches the source fingerprint:

Metric	Description
KS Statistic	Kolmogorov-Smirnov two-sample test statistic (sup of CDF difference).
Wasserstein-1	Earth Mover's Distance via piecewise-linear inverse CDF integration over 9 percentile knots.
JS Divergence	Jensen-Shannon divergence from percentile-bin PMFs, bounded by $\ln 2$.

Distribution CDFs are computed analytically for Normal, LogNormal, Gamma (regularized incomplete gamma via Lanczos + Lentz continued fraction), Pareto, PointMass, and Mixture distributions.

6 Architecture

6.1 Layered Crate Architecture



6.2 Production-Grade Infrastructure

Capability	Details
Deterministic Output	ChaCha8 PRNG with configurable seed; identical seed ⇒ identical dataset.
Financial Precision	`rust_decimal` throughout; no IEEE 754 floating-point artifacts. Decimals serialized as strings.
Resource Guards	Unified CPU, memory, and disk monitoring with automatic throttling and graceful degradation (Normal → Reduced → Minimal → Emergency).
Collision-Free IDs	FNV-1a hash-based UUID factory with generator-type discriminators prevents document ID collisions across parallel generators.
Streaming Output	CSV, JSON, NDJSON, and Apache Parquet (15-column Arrow schema, Zstd compression) streaming sinks with backpressure strategies (Block, DropOldest, DropNewest, Buffer).
API Layer	REST + gRPC + WebSocket with Argon2id API-key auth, JWT/OIDC bearer tokens (RS256, Keycloak/Auth0/Entra ID), RBAC (Admin/Operator/Viewer), sliding-window rate limiting (in-memory or Redis), TLS (rustls), async job queue, security headers, and structured audit logging.
Desktop UI	Tauri + SvelteKit cross-platform application with 15+ configuration pages and real-time streaming viewer.
Plugin SDK	`GeneratorPlugin`, `SinkPlugin`, and `TransformPlugin` traits with thread-safe registry for custom extensions.
Webhooks	Fire-and-forget event dispatch (RunStarted, RunCompleted, RunFailed, GateViolation) with HMAC signing and configurable retry.
Data Lineage	Per-file SHA-256 checksums, `LineageGraph` tracking config→generator→output relationships, W3C PROV-JSON export, CLI `verify` command.
Python Wrapper	`datasynth-py` 1.1 with blueprints (sourcing, HR, manufacturing, financial reporting), pandas/polars integration, async client, and WebSocket streaming support.

7 Industry Presets

DataSynth ships with ten industry-specific configuration presets, each tuned for realistic business process weightings, chart-of-accounts structures, and regional multi-company setups.

Industry	Key Weight	Characteristics
Manufacturing	40% P2P	14 transaction types, multi-level BOM, production routings, work centers, JIT, quality framework.
Retail	50% O2C	12 transaction types, 7 store types, shrinkage modeling, loss prevention, markdown patterns.
Healthcare	15% H2R	15 transaction types, payer mix (Medicare/Medicaid/Commercial), ICD-10/CPT/DRG coding, HIPAA/Stark compliance.
Financial Services	40% R2R	Report-heavy, regulatory compliance, intercompany structures, loan loss provisions.
Technology	15% H2R	SaaS revenue recognition, R&D capitalization, deferred revenue.
Professional Svcs.	—	Time-based billing, project accounting, trust accounting.
Energy	—	Capital-intensive, long-lived assets.
Transportation	—	Fleet management, route-based costing.
Real Estate	—	Property portfolios, lease accounting.
Telecommunications	—	Subscription revenue, network assets.

Each preset supports three complexity tiers: **Small** (∼100 GL accounts), **Medium** (∼400 accounts), and **Large** (∼2 500 accounts).

8 Statistical Distribution Framework

DataSynth includes an advanced statistical distribution framework for generating empirically grounded synthetic data:

Capability	Details
Mixture Models	Gaussian and Log-Normal mixtures with weighted components ("routine", "significant", "major" transactions).
Copula Correlation	Gaussian, Clayton, Gumbel, Frank, Student-t copulas for cross-field dependencies.
Distribution Types	Pareto (heavy-tailed), Weibull (time-to-event), Beta (proportions), Zero-inflated (excess zeros).
Benford's Law	First and second-digit compliance, round number bias, threshold clustering.
Regime Changes	Economic cycles, acquisition/divestiture effects, recession modeling.
Industry Profiles	Pre-configured profiles for Retail, Manufacturing, Financial Services.

All distributions include validation tests: Benford's Law MAD, Anderson-Darling goodness-of-fit, chi-squared, and correlation matrix verification.

9 Use Cases

Primary Use Cases

- **Fraud Detection ML** — Train and validate models on labeled fraud typologies with realistic base-rate imbalance.
- **Graph Neural Networks** — Export transaction graphs in PyTorch Geometric, Neo4j, DGL, or RustGraph format — including multi-layer hypergraphs — with pre-computed features and train/val/test splits.
- **AML & KYC Testing** — Generate banking transaction data with structuring, layering, and mule patterns for compliance system validation.
- **Audit Analytics** — Produce ISA-compliant audit artifacts and anomaly-injected financial data for analytics tool development.
- **ERP Integration Testing** — Generate SAP ACDOCA-format data with full document chains for system migration and integration testing.
- **Process Mining** — OCEL 2.0 event logs for process discovery, conformance checking, and variant analysis.
- **SOX & COSO Compliance Testing** — Internal control definitions with COSO 2013 mappings, SOX 302/404 certifications, deficiency classification, and SoD conflict detection.
- **Accounting Standards Testing** — Generate revenue contracts (ASC 606/IFRS 15), lease portfolios (ASC 842/IFRS 16), fair value measurements, and impairment tests with dual-framework reconciliation.
- **Data Quality ML** — Labeled missing values, typos, duplicates, and format variations for training data-cleansing models.
- **Drift Detection ML** — Labeled drift events (organizational, process, technology, market, regulatory) with magnitude and detection difficulty for drift detection model training.
- **Network Analytics** — Multi-tier vendor networks and customer segmentation graphs with relationship strength metrics for supply chain and customer analytics.
- **Causal Inference Testing** — Structural causal models with do-calculus interventions for counterfactual scenario generation and what-if analysis.
- **Pipeline Orchestration** — Apache Airflow operators, dbt source generation, MLflow experiment tracking, and Spark DataFrame readers for end-to-end ML pipelines.
- **EU AI Act Compliance** — Article 50 synthetic content marking with content credentials and Article 10 data governance reports with bias assessments.

10 Evaluation & Auto-Tuning

DataSynth includes a built-in evaluation framework that measures synthetic data quality across four dimensions:

1. **Statistical Fidelity** — KS statistic, Wasserstein-1 (Earth Mover's Distance), Jensen-Shannon

divergence, Benford's Law MAD, Anderson-Darling goodness-of-fit.

2. **Coherence** — Balance-sheet validation, intercompany matching, document chain integrity, standards compliance.
3. **Data Quality** — Completeness, consistency, duplicate rates, format correctness, uniqueness.
4. **ML Readiness** — Feature distributions, label quality, graph structure, train/val/test split balance.
5. **Drift Detection** — Hellinger distance, Population Stability Index (PSI), rolling window analysis, detection delay metrics.

10.1 ML Benchmarks

Pre-configured benchmarks aligned with ACFE statistics and industry verticals:

- **ACFE-Calibrated**: General fraud (1k), collusion-focused (5k), management override (2k).
- **Industry-Specific**: Manufacturing fraud (5k), retail fraud (10k), healthcare fraud (5k), technology fraud (3k), financial services fraud (5k).
- **Cost-Sensitive**: Asymmetric cost matrices for realistic fraud detection evaluation.

An **auto-tuning engine** analyses evaluation results and produces prioritized configuration patches with expected improvement estimates — closing the loop between generation and validation.

10.2 Quality Gate Engine

New in v0.5.0 — configurable pass/fail thresholds that can block releases of low-quality synthetic data:

- **8 metrics**: Benford MAD, balance coherence, document chain completeness, duplicate rate, missing rate, distribution fit, correlation accuracy, anomaly precision.
- **Built-in profiles**: `strict`, `default`, `lenient` — or define custom per-metric thresholds.
- **CLI integration**: `-quality-gate <profile>` with exit code 2 on failure for CI/CD pipelines.

11 Compliance & Governance

11.1 EU AI Act

DataSynth implements Article 50 (synthetic content marking) and Article 10 (data governance) requirements:

- **Content Credentials**: `SyntheticContentMarker` creates machine-readable credentials (generator name, version, timestamp, config hash) in embedded, sidecar, or both formats.
- **Data Governance Report**: `DataGovernanceReport` with `BiasAssessment` documenting data provenance, quality metrics, and bias evaluation for regulatory submission.

11.2 Regulatory Compliance Documentation

Pre-built compliance guides with self-assessment templates:

Framework	Coverage
EU AI Act	Article 50 content marking, Article 10 governance
NIST AI RMF	Self-assessment across MAP, MEASURE, MANAGE, GOVERN
GDPR	Article 30 records, DPIA guidance, data minimization
SOC 2 Type II	Readiness across 5 Trust Service Criteria
ISO 27001:2022	Annex A alignment (11 implemented, 6 partial)

12 DevOps & Deployment

12.1 CI/CD Pipeline

A 7-job GitHub Actions pipeline ensures release quality:

1. **fmt**: `cargo fmt -check`
2. **clippy**: Lint with `-D warnings`
3. **test**: Cross-platform matrix (Ubuntu, macOS, Windows)
4. **msrv**: Minimum Supported Rust Version validation (1.88)
5. **security**: `cargo deny` + `cargo audit`
6. **coverage**: `cargo-llvm-cov` with Codecov
7. **benchmarks**: Criterion regression checks on PRs

12.2 Deployment Options

Target	Details
Docker	Multi-stage builds with cargo-chef caching; distroless runtime images for server and CLI.
Docker Compose	Server + Prometheus + Grafana stack with auto-provisioned datasources.
Kubernetes	Helm chart with HPA (2–10 replicas), PDB, rolling updates, optional Redis subchart, Prometheus ServiceMonitor, and ConfigMap/Secret templates.
Bare Metal	SystemD service with security hardening (NoNewPrivileges, ProtectSystem, MemoryMax=4G, CPUQuota=200%).
Release Pipeline	Pre-built binaries for 5 platforms, Docker image to GHCR (amd64 + arm64), Trivy container scanning.

12.3 Ecosystem Integrations

Python-native integrations for popular data engineering tools:

- **Apache Airflow**: `DataSynthOperator`, `DataSynthSensor`, and `DataSynthValidateOperator` for orchestrated generation.

- **dbt**: `DbtSourceGenerator` for source YAML and seed files.
- **MLflow**: `DataSynthMlflowTracker` for experiment tracking with quality metrics.
- **Apache Spark**: `DataSynthSparkReader` for direct DataFrame loading from DataSynth output.

13 Getting Started

13.1 Quick Start (CLI)

```
# Generate with demo preset
datasynth-data generate -demo -output ./output

# Create an industry-specific config
datasynth-data init -industry manufacturing -complexity medium -o config.yaml

# Generate from config
datasynth-data generate -config config.yaml -output ./output
```

13.2 Quick Start (Python)

```
from datasynth_py import DataSynth
from datasynth_py.config import blueprints

config = blueprints.retail_small(companies=4, transactions=10000)
synth = DataSynth()
result = synth.generate(config=config)
```

13.3 Server Mode

```
# Start REST/gRPC server with 4 worker threads
cargo run -p datasynth-server - -port 3000 -worker-threads 4
```

Links & Resources

Repository	https://github.com/ey-asu-rnd/SyntheticData
Documentation	https://ey-asu-rnd.github.io/SyntheticData/
Crates.io	https://crates.io/crates/datasynth-core
PyPI	https://pypi.org/project/datasynth-py/

Contact: michael.ivertowski@ch.ey.com